FarmGate Sentinel

EEE 495 Electrical and Electronics Engineering Design II
Project Proposal Presentation

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Motivation: Protect Farm Animals from Predators

Problem. Small/medium farms and ranches face nocturnal predation (e.g., coyotes, wolves, foxes) around paddocks, chicken coops, and barns. Off-the-shelf motion lights/cams often trigger on wind/rain/machinery, don't coordinate with neighbors, and rarely store actionable evidence.

Target Users.

- Small/medium farms, ranches with open paddocks
- Neighbor co-ops benefiting from pooled alerts
- Wildlife managers seeking non-lethal monitoring/deterrence

Why Now? Affordable edge AI + long-range radios enable robust, low-FP detection and coordinated response, and most importantly, harness the power of IoT by combining these into one product.

Proposed Product: FarmGate Sentinel

What it is. A vision-first, multi-node intrusion sensing and deterrence system spanning edge nodes, a site gateway, and a cloud back-end.

How it works on the high level.

- *Edge nodes* (ESP32-CAM/S3) capture periodic frames; burst on suspicion
- *Gateway* (Raspberry Pi 4/5) runs lightweight detection and event formation
- *Cloud* aggregates events across properties, reduces false positives, and pushes alerts
- *Actuators* (spotlight/siren) provide non-lethal deterrence with human-in-the-loop

Value. Reliable alerts, shared situational awareness, and secured evidence with minimal false triggers.

Functions & Related Work

Basic Functions

1. Configurable frame schedule on ESP32-CAM; JPEG upload to gateway
2. Gateway detection and event creation; burst requests on motion/suspicion
3. Cloud correlation across nodes/sites; secure evidence storage; mobile/web alerts
4. Mobile/web UI with incident cards, thumbnails, controls
5. Automated deterrence (spotlight/siren) on confirmed events

Existing Solutions / Related Work

- Lights & sound deterrence: mixed outcomes without policy control **[1,2]**
- IoT farm surveillance with modern detectors (e.g., YOLOv8) **[3]**
- Bioacoustics howl/bark detection; embedded ESC for devices **[4–6]**
- IoT architectures for agriculture (LoRa/LoRaWAN + Pi gateways + cloud) **[7,8]**

System Block Diagram

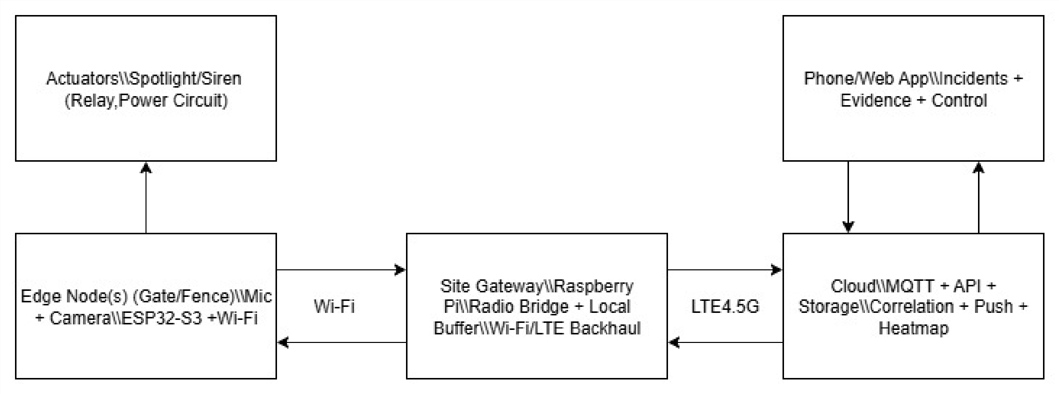

Figure 1: Edge nodes → Site gateway → Cloud & Actuators → Phone/Web App .

Method & Architecture

Approach. Vision-first monitoring at 10 frames/min (policy-adjustable); burst capture during high-risk periods. Secondary audio objective (howl/bark).

Pipeline.

- Edge: ESP32-S3 + camera + (optional) mic/PIR/reed/IMU; tiny-CNN/MFCC (secondary)
- Gateway: Raspberry Pi 4/5; radio bridge; local buffer; lightweight detector; Wi-Fi/LTE backhaul
- Cloud: MQTT + REST/WebSocket; correlation service; object storage; push notifications; RBAC; incident heatmaps

Why WAN? Cross-property fusion, roaming notifications, outage resilience, regional pattern learning.

Equipment / Components

Item	Qty
Raspberry Pi 4/5 (PSU, 32GB)	1
WiFi Module (gateway)	1
USB LTE modem	1
ESP32–S3 development boards	2
Visible-light cameras (Pi Cam v3 / UVC / ESP32-CAM)	2
Digital microphones	2
WiFi modules (edge)	2
Reed switches; PIR sensors,IMU (secondary obj.)	2 sets
Spotlight + compact siren + relay/MOSFET driver	1 set
Cabling, hardware, batteries, custom PCB	—
3D Printed Casings for gateway and edge(s)	—

Objectives, Use Cases & Plan

Objectives. Baseline visual monitoring; reliable animal/person/vehicle detection; incident evidence; cross-property aggregation; automated deterrence with human-in-the-loop.

Representative Use Cases.

- UC-A: Baseline patrol with burst-on-detect and spotlight control
- UC-B: Bandwidth-aware mode with downscaling/skipping near-static frames
- UC-C: (Secondary) Audio-assisted escalation (howl doubles frame rate)
- UC-D: Policy update via cloud (sampling/burst thresholds), essentially giving user(s) control over the process.

Implementation Plan (high level). Research $\rightarrow$ datasets/models/hardware $\rightarrow$ Pi gateway + training $\rightarrow$ cloud skeleton (MQTT/API/push) $\rightarrow$ edge circuit design $\rightarrow$ neighbor fan-out/heatmap $\rightarrow$ mobile app $\rightarrow$ WAN demo $\rightarrow$ tuning/validation.

References

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